

PART A

✓ Question 1: [8 marks]

Consider the following modification made to the Three-Phase Commit (3PC) protocol in a system where at most k processes (including the coordinator) may fail where $k \ll n$. (that is, $n - k > k$)

In this modified version of 3PC, in Phase 2 (the $\langle \text{PreCommit} \rangle$ phase), the coordinator does not wait for acknowledgements from all processes before proceeding. Instead, the coordinator waits until it has received acknowledgements from k processes in response to its $\langle \text{PreCommit} \rangle$ message. The value of k is fixed by the user. We assume that more than k processes never fail. As soon as the coordinator receives these k acknowledgements, it begins sending $\langle \text{Commit} \rangle$ messages to those processes that have acknowledged the $\langle \text{PreCommit} \rangle$ message. Furthermore, the coordinator continues to send $\langle \text{Commit} \rangle$ messages to any additional processes that send acknowledgements for $\langle \text{PreCommit} \rangle$ at a later time.

This differs from the original 3PC protocol, in which the coordinator waits for acknowledgements from all processes before sending any $\langle \text{Commit} \rangle$ messages. All other assumptions of the original 3PC protocol remain unchanged.

- ✓ A. Does this modified version preserve the non-blocking properties of 3PC? [4 marks]
- ✓ B. Can this be considered a correct or acceptable variation of 3PC? [4 marks]

✓ Question 2: [8 marks]

Consider the following three scenarios in 3PC

Scenario 1: In phase 2 the coordinator C sends preCommit messages to all cohorts and has gotten an ack from all except cohort A. Network problems prevent cohort A from receiving the coordinator's preCommit message. Node A times out waiting for the preCommit message and chooses to abort. Then both the coordinator and node A crash.

Scenario 2: The protocol reaches phase 3. The coordinator sends a doCommit message to only node A. Other nodes have not received the doCommit message. Before it can send more doCommit messages the coordinator crashes. Node A commits its part of the transaction then crashes.

- ✓ A. In Scenario 1: Mention and explain if you notice any inconsistency in the decision taken by Node A vs the other nodes. Does this indicate that 3PC is an incorrect commit protocol [4 marks]
- ✓ B. In Scenario 2: how will the remaining nodes proceed after the coordinator and Node A fail. [4 marks]

✓ PART B [15 marks]

Consider Raymond's algorithm for distributed mutual exclusion. Answer the following:

- ✓ 1. Provide an intuitive but logically grounded explanation on Raymond's algorithm being starvation-free under the assumptions of reliable communication channels, no process crashes, and no link failures. [2 marks]
- ✓ 2. Analyse whether the algorithm guarantees global First-Come-First-Served (FCFS) ordering of requests across all nodes. Discuss your reasoning. [3 marks]
- ✓ 3. Identify potential issues if nodes can crash. What aspects need special consideration if you were to modify the algorithm to handle node crashes (where the node once crashed will not come back)? Provide a rough outline of your approach to incorporate fault-tolerance for node crashes. [5 marks]
- ✓ 4. Discuss the problems that may arise due to link or channel failures. What aspects need special consideration if you were to modify the algorithm to handle link failures? Provide a rough outline of your approach to incorporate fault-tolerance for link failures. [5 marks]

Note: You may assume reasonable system properties, such as bounded message delivery times or limited crash recovery time. You may also refer to standard algorithms (e.g., leader election) or concepts without providing detailed descriptions.